

HOBOKSAR, China

WMO: 511560

Lat: 46.8103N

Lon: 85.7498E

Elev: 4341

StdP: 12.53

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-5.9	-20.0	2.2	-7.1	-16.2	2.7	-2.4	23.3	14.1	18.0	15.3	3.1	320	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	85.0	58.6	82.3	57.5	79.6	56.7	61.2	77.1	60.0	75.9	58.8	74.4	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	78.4	66.2	54.1	73.3	65.9	52.3	68.3	65.9	29.4	76.9	28.4	76.1	27.5	74.4	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.2	18.8	14.9	DB	-13.7	89.8	4.8	3.4	-17.1	92.2	-19.9	94.3	-22.6	96.2	-26.0	98.7
(5)				WB	-14.2	63.4	4.6	2.2	-17.6	65.0	-20.3	66.2	-22.9	67.4	-26.2	69.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	9.8	15.6	28.9	44.4	55.1	65.0	68.0	65.2	54.6	41.0	25.7	13.9
	DBStd	21.96	7.85	8.56	10.27	9.07	7.80	6.05	5.31	5.83	7.83	8.69	9.60	7.69
	HDD50	5298	1246	964	653	206	40	1	0	1	42	297	730	1118
	HDD65	9145	1711	1384	1118	618	315	73	29	69	320	745	1180	1583
	CDD50	1915	0	0	0	38	199	450	557	473	181	17	0	0
	CDD65	286	0	0	0	0	9	72	120	76	9	0	0	0
	CDH74	2346	0	0	0	6	100	556	967	635	82	0	0	0
	CDH80	518	0	0	0	0	10	106	255	135	12	0	0	0
Wind	WSAvg	5.2	3.2	3.7	5.0	7.0	7.5	6.4	6.0	5.8	5.8	5.0	4.1	3.3
Precipitation	PrecAvg	5.4	0.1	0.1	0.2	0.3	0.5	1.1	1.5	0.7	0.4	0.2	0.2	0.1
	PrecMax	8.2	0.5	0.4	0.8	0.7	1.2	3.0	2.9	2.0	1.3	0.6	0.7	0.5
	PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.2	0.0	0.0	0.0	0.0
	PrecStd	1.5	0.1	0.1	0.2	0.2	0.3	0.6	0.8	0.4	0.4	0.2	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	38.8	59.1	72.9	81.0	86.3	90.5	87.6	81.6	68.0	52.2	35.3
		MCWB	28.7	30.5	41.9	50.0	54.4	58.5	60.9	59.2	55.6	47.9	41.0	29.2
	2%	DB	27.6	34.2	53.3	67.0	76.8	83.1	85.9	83.7	75.8	62.5	47.7	31.1
		MCWB	23.5	27.8	39.5	46.9	52.7	57.4	59.8	57.7	53.2	45.2	38.5	26.1
	5%	DB	23.9	31.1	48.8	62.8	73.0	80.2	82.7	80.6	72.0	58.4	43.2	27.6
		MCWB	20.2	25.5	37.5	44.6	51.2	56.5	58.3	57.1	51.6	43.8	35.4	23.5
	10%	DB	20.8	27.8	44.2	58.9	69.2	77.0	79.5	77.4	68.4	54.4	38.4	24.5
		MCWB	17.7	23.0	35.1	43.6	49.5	55.6	57.5	56.3	50.2	42.1	32.1	21.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.3	31.8	43.8	52.1	56.8	61.0	64.5	62.8	56.7	49.4	42.7	29.9
		MCDB	33.3	36.7	56.3	71.0	75.5	78.1	82.2	77.5	75.3	64.3	49.8	34.7
	2%	WB	24.1	28.4	40.6	48.8	54.5	59.7	62.1	60.5	54.8	46.8	38.9	26.6
		MCDB	27.2	33.5	51.6	63.1	71.6	76.5	77.6	77.6	72.2	59.1	46.7	30.2
	5%	WB	20.5	25.6	38.2	46.5	52.9	58.4	60.7	59.1	53.1	44.8	35.5	23.8
		MCDB	23.6	30.6	47.9	59.2	69.4	74.8	75.8	75.0	68.9	56.4	42.7	27.3
	10%	WB	17.7	23.1	35.6	44.5	51.2	57.2	59.4	57.8	51.2	42.7	32.5	21.2
		MCDB	20.8	27.5	43.6	56.9	66.5	73.4	74.6	73.4	66.5	53.3	38.0	24.4
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.8	17.5	18.8	19.0	18.0	17.8	19.3	19.5	17.7	14.3	14.4
		MCDBR	16.7	18.5	21.0	23.4	22.9	21.5	21.9	22.7	23.2	22.3	18.4	16.6
		MCWBR	13.5	13.8	11.9	11.5	9.8	8.0	7.4	8.6	10.3	11.8	11.7	13.0
	5% WB	MCDBR	16.1	17.6	19.7	19.3	19.6	18.4	17.7	19.1	20.6	20.1	17.6	15.9
		MCWBR	13.1	13.3	11.8	10.3	9.0	7.5	6.9	7.8	9.7	11.2	11.9	12.8
Clear-Sky Solar Irradiance	taub	0.269	0.298	0.339	0.386	0.374	0.387	0.386	0.363	0.334	0.321	0.292	0.270	
	taud	2.211	2.153	2.150	2.136	2.243	2.272	2.307	2.353	2.387	2.330	2.301	2.219	
	Ebn at Noon	259	273	278	274	282	278	277	279	279	263	248	242	
	Edh at Noon	26	34	40	45	42	41	39	36	32	29	24	23	
All-Sky Solar Radiation	RadAvg	567	897	1289	1688	2089	2164	2077	1898	1516	942	530	411	
	RadStd	44	47	91	96	112	112	137	90	86	77	55	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	-0.76	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page